

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Ca $\times$ $\times$ $\rightarrow$ $\left[\begin{array}{c} \cdot \cdot \\ \cdot \text{F} \cdot \\ \cdot \cdot \end{array} \right]^-$ 2,8 $\downarrow$ $\rightarrow$ $\left[\begin{array}{c} \cdot \cdot \\ \cdot \text{F} \cdot \\ \cdot \cdot \end{array} \right]^-$ 2,8 $\left[\text{Ca} \right]^{2+}$ 2,8,8 award (1) for <u>transfer of electrons</u> from calcium atom to two fluorine atoms award (1) for electronic structure and charge of calcium ion award (1) for electronic structure and charge of one fluoride ion award second/third mark if both electronic structures given (no/incorrect charges) or both charges given (no/incorrect electronic structures) accept diagrams with all electron shells shown		3		3		